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$$\frac{2(2n+3)}{n(n+1)(n+2)(n+3)} = \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2} + \frac{D}{n+3}$$

$$A = \frac{6}{6} = 1$$

$$B = \frac{2}{-2} = -1$$

$$C = \frac{-2}{2} = -1$$

$$D = \frac{-6}{-6} = 1$$

$$\frac{2(2n+3)}{n(n+1)(n+2)(n+3)} = \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3}$$

Eerste 4 termen opschrijven:

$$s_1 = \frac{1}{1} - \frac{1}{2} - \frac{1}{3} + \frac{1}{4}$$

$$s_2 = \frac{1}{2} - \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$$

$$s_3 = \frac{1}{3} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6}$$

$$s_4 = \frac{1}{4} - \frac{1}{5} - \frac{1}{6} + \frac{1}{7}$$

$$s_{n-1} = \frac{1}{n-1} - \frac{1}{n} - \frac{1}{n+1} + \frac{1}{n+2}$$

$$s_n = \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3}$$

Bij optellen zien wij duidelijk dat bijna alle termen afvallen behalve:

$$\begin{aligned} &\Rightarrow 1 - \frac{1}{3} - \frac{1}{n+1} + \frac{1}{n+3} \\ &= \frac{3(n+3)(n+1) - (n+3)(n+1) - 3(n+3) + 3(n+1)}{3(n+3)(n+1)} \\ &= \frac{2(n+3)(n+1) - 3(n+3) + 3(n+1)}{3(n+3)(n+1)} \\ &= \frac{2n^2 + 2n + 6n + 6 - 3n - 9 + 3n + 3}{3(n+3)(n+1)} \\ &= \frac{2n^2 + 8n}{3(n+3)(n+1)} \end{aligned}$$

References